Native inherited-limit gate — source proposal

The candidate adds a native receipt and fail-closed AS/DATA qualification before the first findmnt or Git subprocess. It supersedes the manual post-launch limit-observation gap in the inert additional-admission proposal. No native execution, compiler, tests, imported candidate or resource limit mutation occurred. PCC unverified; independent source review pending.

Concrete policy and source order

After the existing owned-output and physical-cgroup checks, the worker checkpoints phase native-inherited-limit-qualification and an incomplete native_limits record. It reads /proc/self/limits as bytes, retains the decoded text, byte count and SHA256, and flushes that receipt. It obtains resource.getrlimit(RLIMIT_AS) and resource.getrlimit(RLIMIT_DATA), retaining exact soft/hard values and their comparisons with resource.RLIM_INFINITY. It checkpoints qualified only when all four values equal infinity; otherwise it records rejected and raises before any child command. Read, decoding, getrlimit or durable checkpoint failures propagate to the existing failure/terminal receipt path and cannot reach findmnt. If output storage itself fails, complete receipts may remain unavailable, but no child is launched from this gate.

This conservative policy uses no guessed mapping-size allowance and never calls setrlimit. It can reject an environment whose finite limits would actually suffice. Such a rejection must be retained; changing the policy requires separate review, not a silent retry. Unlimited AS/DATA does not imply sufficient resident memory or successful mapping: the unchanged hard4GiB cgroup, kernel mapping constraints and other failures remain. Snapshot observations occur at gate time; external privileged mutations after the snapshot are outside this source invariant. The policy is enforced by an explicit conditional raise, not by a Python assert. Existing admission guards retain their original assert semantics and exact published python invocation has no optimization flag.

Receipt compatibility and preserved behavior

The historical native receipt keys status, revision, compiler_invoked, errors, files, phase, elapsed_seconds, cgroup and child_cleanup remain. native_limits is additive and contains policy, pid, status, limit_mutation_performed=false, proc_self_limits_utf8, proc_self_limits_sha256, proc_self_limits_bytes, getrlimit and rlim_infinity_constant when available. A partially filled record means incomplete observation. A qualified record is not materialization success or ownership proof. Existing final errors, owned-group cleanup, manifest custody and status checks remain required.

The source diff adds only this gate after cgroup checkpoint and before the existing first cmd call. The helper is byte-identical. Source/input pin29861c9a55f9ae5a33e1e09b3866aa397fbde605 and candidate-pencil555a5c62339a927579599d46c7e94265e52497dc56bacf0b13bb53c2dcd133c4 remain. Existing native880seconds and outer900seconds,2CPU,4GiB,zero swap,512tasks,one owner/no overlap and no automatic retry remain. There is no compiler in this stage.

Inert configuration and accounting

The successor config and command use new prospective paths/unit, the new candidate hash and unchanged helper. approval_required=true and launch_disabled=true remain; reviewed=false, source review commit=null, publication=null and runtime lease=null. The unmodified admission branch rejects before owned output. Do not use the source-preparation task lease as runtime authorization. This packet is not an additional attempt approval. Prior1/1 remains consumed; total46800 minus4500charged leaves42300. Proposed900 preparation and1800 subsequent validation are unchanged arithmetic; no new worker debit was incurred here. Deadline stays2026-11-06 04:00UTC. An admitted successor must be independently reviewed, fully bound and published after explicit additional-attempt authorization. No materialized-input manifest or native limit values are fabricated.

Source reasoning and review boundary

All subprocess creation is through cmd. In the existing linear try-body the new qualification precedes both initial findmnt commands and later Git calls; no cmd call occurs earlier. Exceptions leave that body for the existing failure/finalization path. The new checks record this process's values, avoiding the external /proc/PID observation race. A successful native gate establishes only the stated source precondition before child creation, not full runtime correctness. Static inspection only; independently review the exact candidate.patch, result.json hashes and expected source-order.csv before any prospective run. Prior mapping review applies to the parent source, not this new gate.

Sources checked: frozen dbe4d4f9 materializer; prior receipt and cleanup; inert proposal714e95ca; mapping reviewd7222eee; existing reviewer finding on missing native qualification. KB tools/wrapper remain unavailable; exact repository source and prior primary getrlimit documentation are the fallback. No new mathematical or novelty claim.

Native inherited-limit gate: source proposal, unexecuted

Owned output	Physical caps	Native AS/DATA	Checkpoint policy	First child cmd

Finite or unreadable native limits reject before child commands; no limit is raised.